

IMEN266 Operations Research II — Homework 3 (Ch. 5, exponential distribution, Poisson process, CTMCs)

Fall 2026

Not collected, not graded — but: **some Part-A problems may reappear on the exams.**
Work through it with the companion notebook `HW3.ipynb`; Part-A solutions follow on PLMS.

AI rules for this homework (details: `docs/ai.usage.guide.pdf`)

- **Part A:** solve *alone first*, closed-book style, on paper. Only afterwards may you check with an AI — and if the AI changes your mind, record why in Part D.
- **Parts B–C:** AI collaboration is expected. You remain responsible for every claim you make.
- **Part D:** keep the prompt log whenever AI was used — it is your own audit trail (the homework is not collected).

Part A — Exam-style warm-up (no AI on first attempt)

- A1. (**Exponential distribution, Poisson process.**) Calls reach a help desk according to a Poisson process with rate 6 per hour. (i) The probability of no call in the next 15 minutes. (ii) The probability of at least 3 calls in the next 30 minutes. (iii) No call has arrived in the last 10 minutes. What is the expected waiting time until the next call, and the probability that it arrives within 5 minutes? Justify in one sentence. (iv) Independently of everything else, 40 % of the calls are complaints. What is the probability of exactly 2 complaint calls in the next hour? (v) A second desk receives its own calls as an independent PP(4) per hour. Consider the combined stream: what is the probability that the next call (of either desk) arrives within 5 minutes *and* is for the first desk? What is the expected number of second-desk calls between two consecutive first-desk calls?
- A2. (**Modeling.**) A walk-in clinic has 2 doctors and 2 waiting chairs (at most 4 patients in the clinic). Patients arrive as PP(λ); each doctor treats one patient at a time for an $\exp(\mu)$ time. An arriving patient who finds both chairs occupied leaves immediately; one who finds exactly one chair occupied joins with probability $\frac{1}{2}$ (and leaves otherwise); one who finds a free doctor or an empty waiting room always joins. A *waiting* patient (not yet with a doctor) loses patience after an $\exp(\theta)$ time and leaves. Let $X(t)$ be the number of patients in the clinic at time t . Model $\{X(t), t \geq 0\}$ as a CTMC: state space, generator Q (with the reasoning behind every rate — which exponential clocks are running?), rate diagram. Is it a birth & death process?
- A3. (**Limiting distribution, cost.**) A workstation is *blocked* (state 0), *working* (state 1) or *starved* (state 2), with generator (time unit: hours)

$$Q = \begin{pmatrix} -4 & 3 & 1 \\ 2 & -3 & 1 \\ 1 & 1 & \square \end{pmatrix}.$$

- (i) Fill the blank and give the mean sojourn time in each state. (ii) Find the limiting distribution $\mathbf{p}$ (exact fractions). (iii) Give the embedded (jump) chain P and its stationary distribution π ; check $p_j \propto \pi_j / \lambda_j$. (iv) Blocking costs 100 (thousand won) per hour and starvation 60 per hour;

working costs nothing. Compute the long-run average cost per hour. (v) What is the mean time between two successive *entries* into the blocked state? (Hint: in the long run, entries into 0 occur at rate $\lambda_0 p_0$.)

- A4. **(Transient analysis.)** A machine has $\exp(\mu)$ up times with mean 5 days and $\exp(\lambda)$ repair times with mean 1 day. (i) Write Q and, using the closed form of $P(t)$ for a two-state CTMC (Companion Proof 5), compute $P(\text{up at } t = 2 \mid \text{down at } 0)$ and $P(\text{down at } t = 2 \mid \text{up at } 0)$. (ii) What is the long-run availability, and after how many days is $P(\text{up at } t \mid \text{down at } 0)$ within 1 % of it? (iii) Estimate the expected number of failures in a year. (Hint: failures occur at rate μ while the machine is up.) (iv) The embedded chain has $\pi = (\frac{1}{2}, \frac{1}{2})$. Explain in two sentences why this does not contradict (ii).

Part B — Critique an AI solution (find the flaws)

A student pasted the repair-person problem of CP8 #9 with numbers — two identical machines, mean up time 4 days, mean repair time 1 day, revenue $r = 500$ per up machine per day, charge $c = 300$ per working repair person per day — into an AI assistant and asked:

“Should we employ one or two repair persons? Also: if both machines are down now, what is the probability that both are up one day later, and what is the probability that a working machine fails within the next 3 days?”

The AI replied:

Step 1. Let $X(t)$ be the number of machines up, $S = \{0, 1, 2\}$, $\mu = 1/4$, $\lambda = 1$. With one repair person the generator is $Q = \begin{pmatrix} -1 & 1 & 0 \\ 0.25 & -1.25 & 1 \\ 0 & 0.5 & -0.5 \end{pmatrix}$.

Step 2. The long-run fractions of time follow from the jump chain

$P = \begin{pmatrix} 0 & 1 & 0 \\ 0.2 & 0 & 0.8 \\ 0 & 1 & 0 \end{pmatrix}$: solving $\pi = \pi P$ gives $\pi = (0.1, 0.5, 0.4)$, so the machines spend 10% of the time both down, 50% with one up and 40% with both up.

Step 3. With two repair persons the jump chain is exactly the same matrix P , so the time fractions are the same; the second person only adds cost. Reward with one person: $-300(0.1) + 200(0.5) + 1000(0.4) = 470$ per day; with two: $-600(0.1) + 200(0.5) + 1000(0.4) = 440$. Employ one repair person.

Step 4. Cross-check with $Q\mathbf{p} = 0$: $Q(1, 1, 1)^\top = 0$, so the uniform distribution $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is also stationary. The chain has several stationary distributions; I keep the one from Step 2.

Step 5. Transient: $P(1) \approx I + Q$, so $P(X(1) = 2 \mid X(0) = 0) = (I + Q)_{02} = 0$.

Step 6. A working machine fails at rate $\mu = 0.25$ per day, so the probability that it fails within 3 days is $0.25 \times 3 = 0.75$.

Your tasks:

- B1. Identify **at least three distinct errors** in this reasoning. Name each precisely (which distribution is confused with which; which equation is transposed; which approximation is misused and for which range of t it would be acceptable; which quantity is not a probability). *One error hides another: explain why “the jump chain is the same for one and two repair persons” is true and nevertheless leads to a wrong decision.*

- B2. Write the correct solution: p for one and for two repair persons (arc balance, exact fractions), both long-run rewards and the decision (and the general rule $r/\mu \geq c/\lambda$), $P(X(1) = 2 \mid X(0) = 0)$ from e^Q , and the 3-day failure probability. You may use HW3.ipynb for e^Q , but the formulas must be on paper.
- B3. Verify the two rewards by simulating the two systems *directly with competing exponential clocks* (a skeleton is provided; do not use the generator). Report the estimates with 95 % confidence intervals over independent replications.

Part C — Stress-test the assumption (AI + simulation)

Every model in this chapter rests on exponential clocks. Suppose the *means* are right but the *shapes* are not: repair times are Erlang or even deterministic, up times are lognormal. Which answers survive?

- C1. **Before simulating**, discuss with your AI assistant and write a one-line prediction (with a reason) for each of the following. Mean up time 2 days, mean repair time 1 day throughout.
 (a) One machine, one repair person: does the long-run availability change when the repair-time distribution changes (same mean)? When the up-time distribution changes? (b) Two machines, one repair person (CP8 #9, case 1): does the time fraction with both machines down change when the repair times become deterministic? When the up times become deterministic (repair still exponential)? (c) Two machines, two repair persons (case 2): same two questions.
- C2. Simulate all cases with the event-driven simulator provided in HW3.ipynb (it accepts any distribution for up and repair times) and compare with the CTMC values. Which of your predictions failed? Ask your AI assistant for the name and the statement of the result that explains the asymmetry you observe in (b), and summarise it in at most five sentences (you are not expected to prove it). *Moral: “exponential” is an assumption about shape, and some answers depend on shape while others depend only on means.*
- C3. (Optional) The Erlang loss formula (CP8 #3) is claimed to be *insensitive* to the distribution of the call durations. Test it: run the K -channel switch with deterministic and with lognormal durations of the same mean and compare the fraction of lost calls with $B(K, \lambda/\mu)$. Then replace “lost” by “queued” (an $M/G/1$ queue) and show that the mean number in the system is *not* insensitive.

Part D — Reflection and prompt log (keep it whenever AI was used)

Fill the table in HW3.ipynb (or reproduce it in your PDF): one row per meaningful AI interaction — where you used it, the prompt, what it returned, and **what you checked or corrected**. Then answer in 2–3 sentences: *where was the AI most useful, and where was it least trustworthy, on this homework?*

Logistics. This homework is not collected. Solutions to Part A will be posted on PLMS about 10 days after release. Parts B–D have no single “answer key”; bring your questions to class or office hours.